

CPSC 317 – Summer 2026Summer
2026

Introduction to Computer Networking

Network Performance

Module 2.1

Ibtissem

READING

- Reading: 1.4.4

Learning Goals

Basic Definitions

- Define the terms bandwidth, throughput, goodput, latency, round-trip-time, and jitter.
- Explain the difference between bandwidth, latency, throughput and goodput
- Perform computations with respect to bandwidth, throughput, goodput, and latency
- Calculate the protocol overheads with respect to performance (goodput vs bandwidth)
- Describe how jitter is introduced into a network

Network Metrics

- Bandwidth
- Throughput
- Goodput
- Latency
- Round-trip-time
- Jitter

Network Metrics: Bandwidth

- Bandwidth: maximum rate at which data can be sent over a link
 - When you see an advertised rate, it is almost always bandwidth
 - 1 Gbps ethernet
 - 500 Mbps or 1 Gbps “speed” from Shaw
 - 2,500 Mbps “speed” from Telus

Units of Measurement in this Course

- For simplicity, we will use the following types of notation throughout this course:
- **Data Size:** [K/M/G][b/B], e.g. Kb, GB, KB
 - In this case, K, M, G, T follow powers of 2:
 - $K = 2^{10}$, $M = 2^{20}$, $G = 2^{30}$
- **Data Rates:** [K/M/G] bps, e.g. Kbps, Gbps
 - In this case, K, M, G, T follow powers of 10:
 - $K = 10^3$, $M = 10^6$, $G = 10^9$
- **Reminder:** 1 Byte (B) = 8 bits (b)

Data Size Vs Data Rate

- Discrepancy grows with size
 - 10^3 vs 2^{10} off by 2.4%
 - 10^6 vs 2^{20} off by 4.9%
 - 10^9 vs 2^{30} off by 7.4%
 - 10^{12} vs 2^{40} off by 10.0%

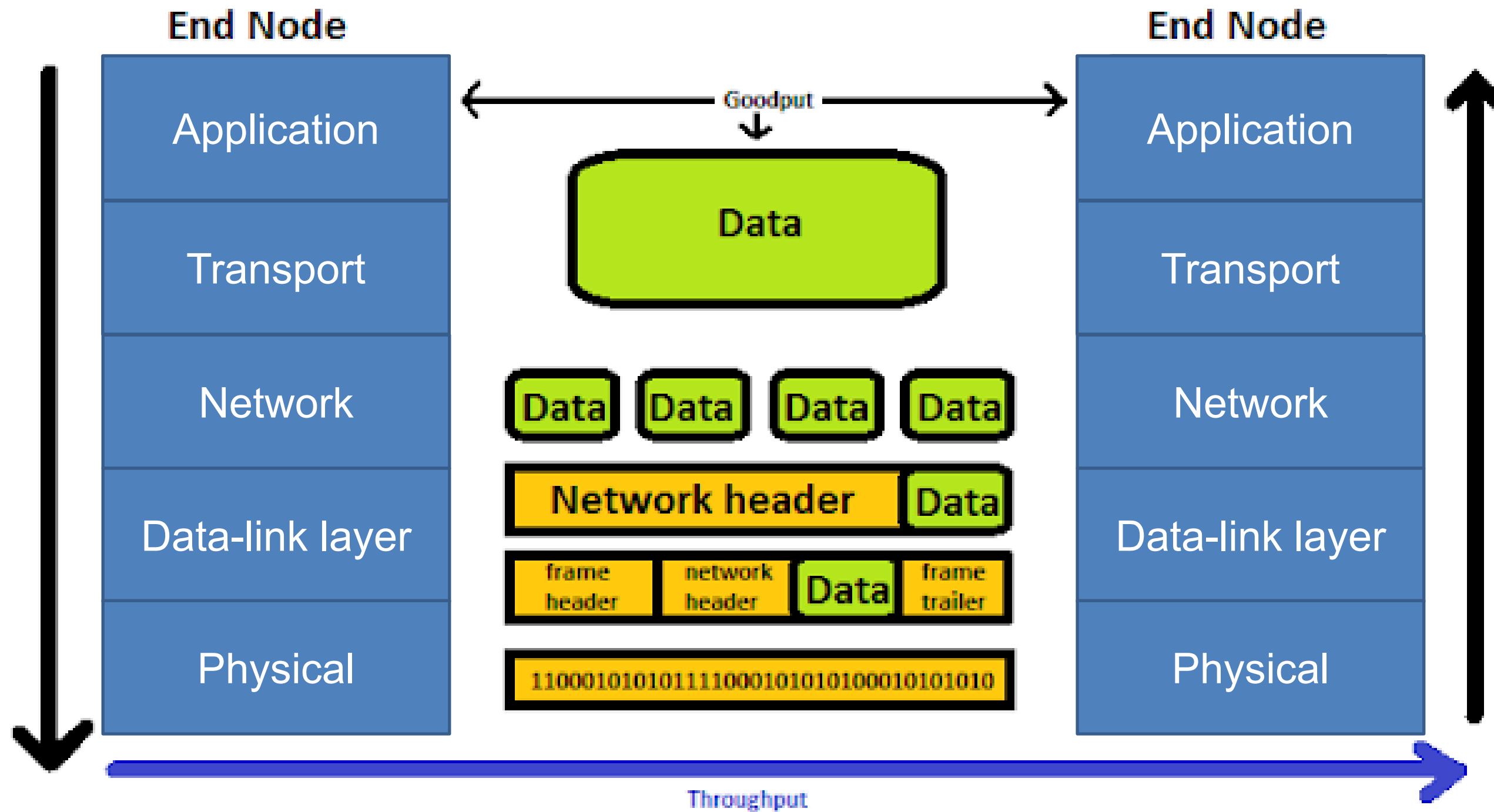
Network Metrics: Throughput

- **Throughput:** amount of data moved from one location to another in a given time
- Usually expressed in bits per second (bps, Mbps, etc.)

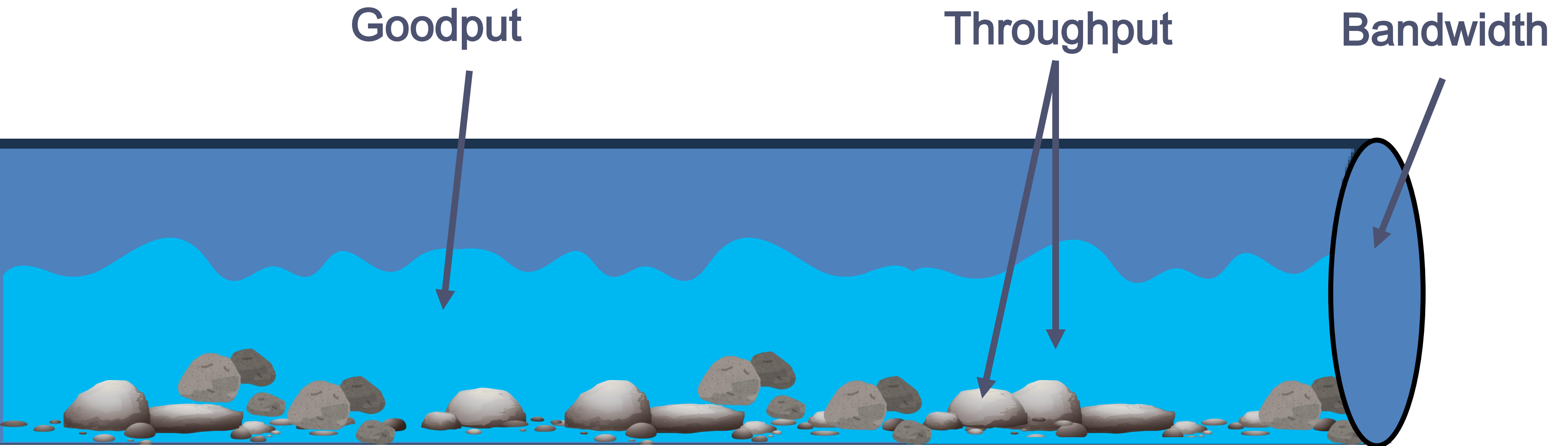
Network Metrics: Goodput

- **Goodput:** rate at which *useful* data arrives
 - Does not include headers and encoding costs
 - Does not include data loss and retransmissions
- May depend on context and application-layer protocol

Goodput Vs Throughput



Goodput Vs Throughput Vs Bandwidth



Goodput Vs Throughput Vs Bandwidth



Clicker Question

A file of size 10 MB is being transferred over a network connection. The network operates at a data rate of 8 Mbps (Megabits per second). The total transmission time for the file is 12 seconds. During the transmission, 1 MB of the data consists of protocol overhead (headers, acknowledgments, retransmissions, etc.) is also sent.

Calculate the Throughput and Goodput in Mbps.

Clicker Question

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Calculate the Throughput and Goodput in Mbps.

Clicker Answer

Throughput = Overall data sent / Time required to send data

Overall data sent = file size + overhead = 10+1 = 11 MB = $11 * 8 * 2^{20}$ bits

Time required to send data = 12 s

Throughput = $11 * 8 * 2^{20} / 12 = 7\,689\,557.33$ bps = **7.6895 Mbps**

Goodput = Useful data sent / Time required to send data

Goodput = $10 * 8 * 2^{20} / 12 = 6\,990\,506.66$ bps = **6.99 Mbps**

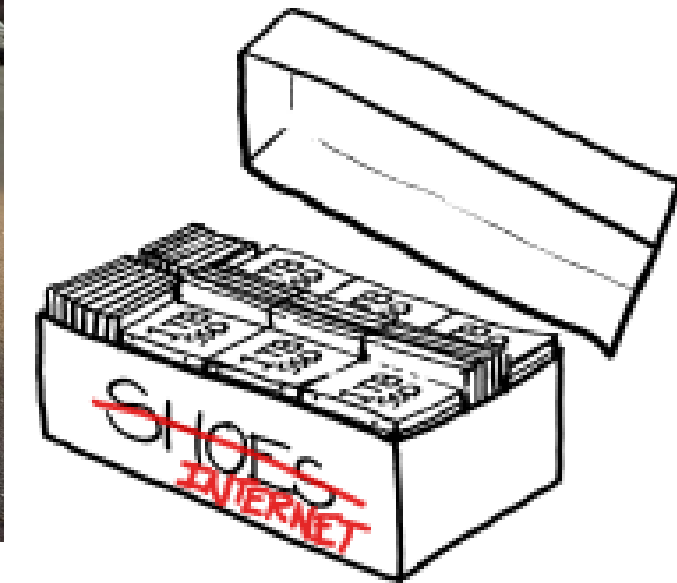
Bandwidth and Station-Wagons

“Never underestimate the bandwidth of a station wagon full of tapes hurtling down the highway.”—Andrew Tanenbaum, 1981



Cisco **estimates** that total **internet** traffic currently averages **167 terabits per second**. **FedEx** has a fleet of 654 aircraft with a **lift capacity of 26.5 million pounds daily**. A solid-state laptop drive weighs **about 78 grams** and can hold up **to a terabyte**.

That means **FedEx** is capable of transferring **150 exabytes** of data per **day**, or **14 petabits per second**—almost a **hundred times** the current throughput of the **internet**.



TOP-END LAPTOP DRIVES: 136
STORAGE: 136 TERABYTES
COST: \$130,000
(PLUS \$40 FOR THE SHOES)



MICROSD CARDS: 25,000
STORAGE: 1.6 PETABYTES
RETAIL COST: \$1.2 MILLION

SUV vs Fiber

The distance of 413km from UBC-V to UBC-O can be covered at an average speed of 75km/hr in an SUV with 3.5m^3 of carrying capacity. A Blu-Ray in a jewel case takes up 88.75 cm^3 . There is a 100 Gbps data connection between the two institutions. You have 260,000 GBytes of data, currently stored on Blu-Ray discs that you want to get to UBC-O and store on disk there. Each disc, on average, has 40 GBytes of data. Ignore van loading and unloading costs and reading/writing disc costs (i.e., the time costs of duplicating a Blu-ray Disc to send it). Consider the 100 Gbps data connection as goodput

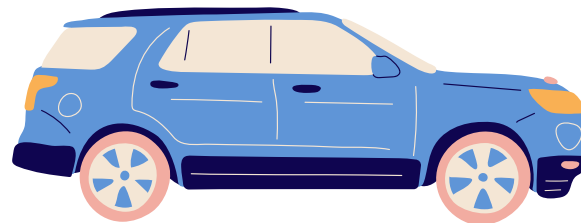
SUV vs Fiber

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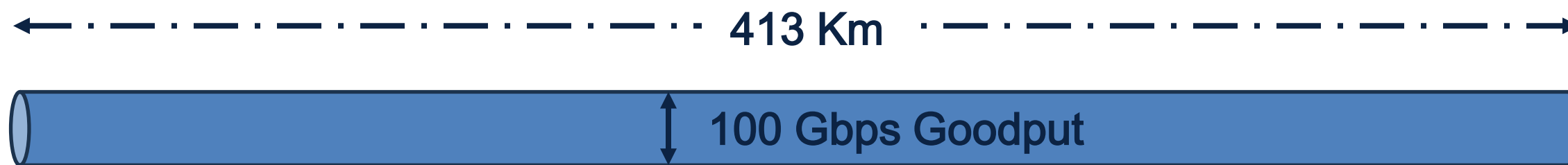
SUV vs Fiber

3.5m³ of
carrying
capacity

75 Km/h



VANCOUVER



OKANAGAN

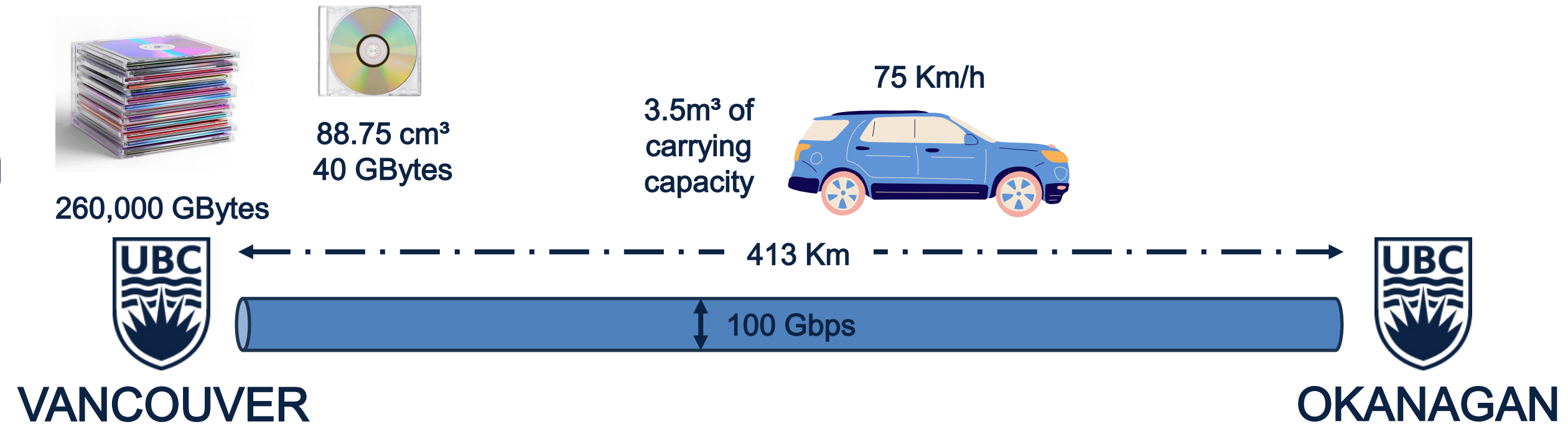


260,000 GBytes



88.75 cm³
40 GBytes

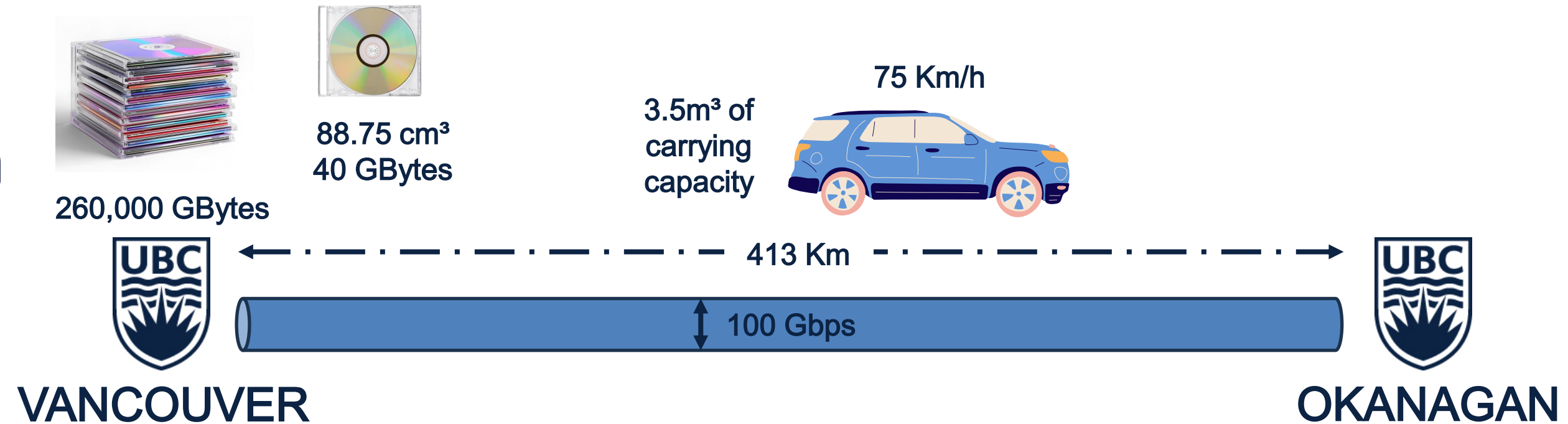
Clicker Question



Without doing a detailed calculation, predict which will get the data to UBC-O faster?

- A. SUV
- B. Network connection

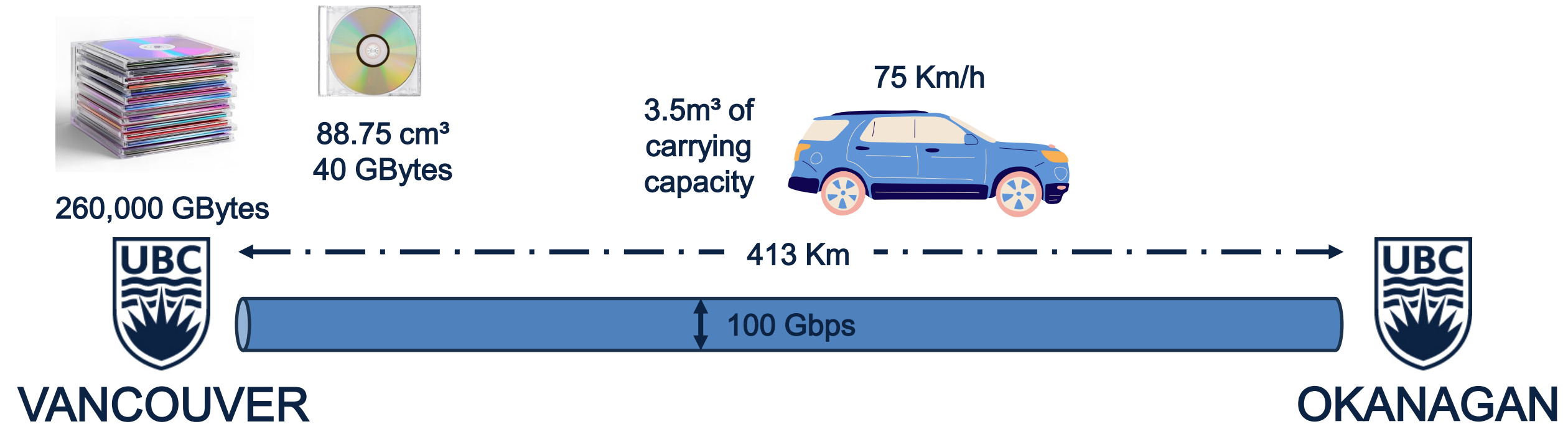
Clicker Question



After 180 minutes, how many gigabytes of data would have arrived at UBC-O by network connection ?

- A. 0 GB
- B. 126 000 GB
- C. 1 080 000 GB

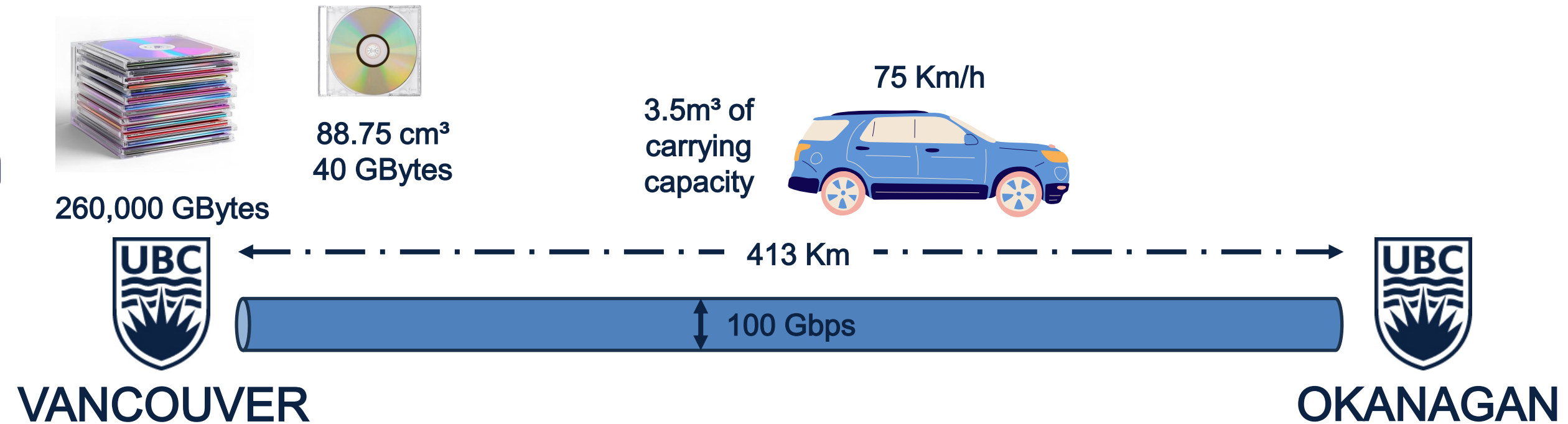
Clicker Answer



After 180 minutes, how many gigabytes of data would have arrived at UBC-O by network connection ?

- A. 0 GB
- B. 126 000 GB ($180 \times 60 \times 100 \times 10^9 = 1.08 \times 10^{15}$ bits / 8 = 1.35×10^{14} Bytes / $2^{30} \approx 126\,000$ GB)**
- C. 1 080 000 GB

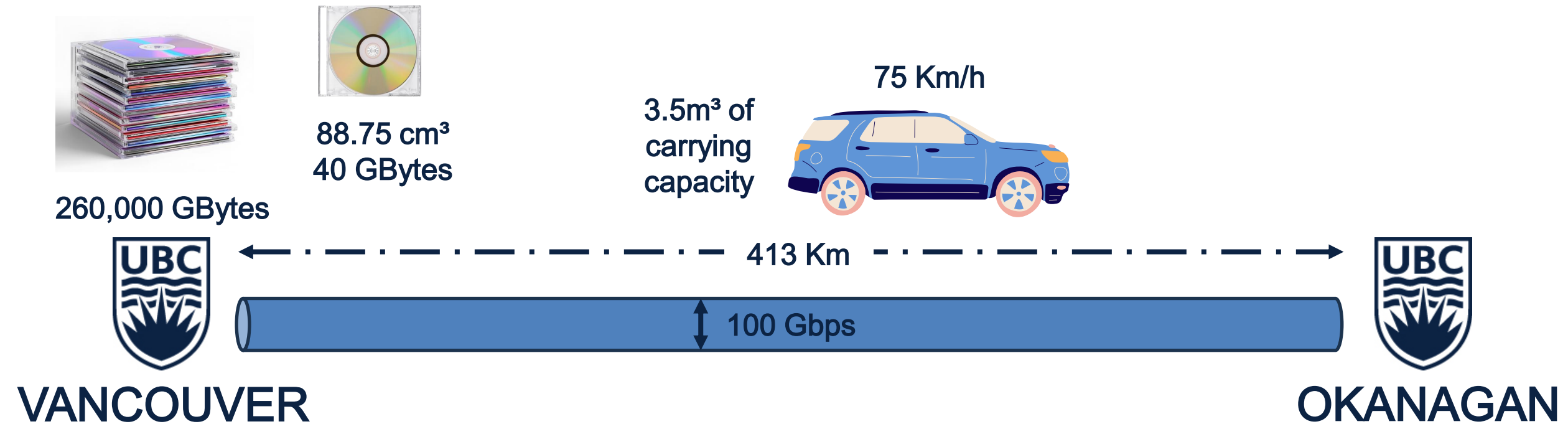
Clicker Question



After 180 minutes, how many gigabytes of data would have arrived at UBC-O by SUV?

- A. 0 GB
- B. 135 000 GB
- C. 1 080 000 GB

Clicker Answer



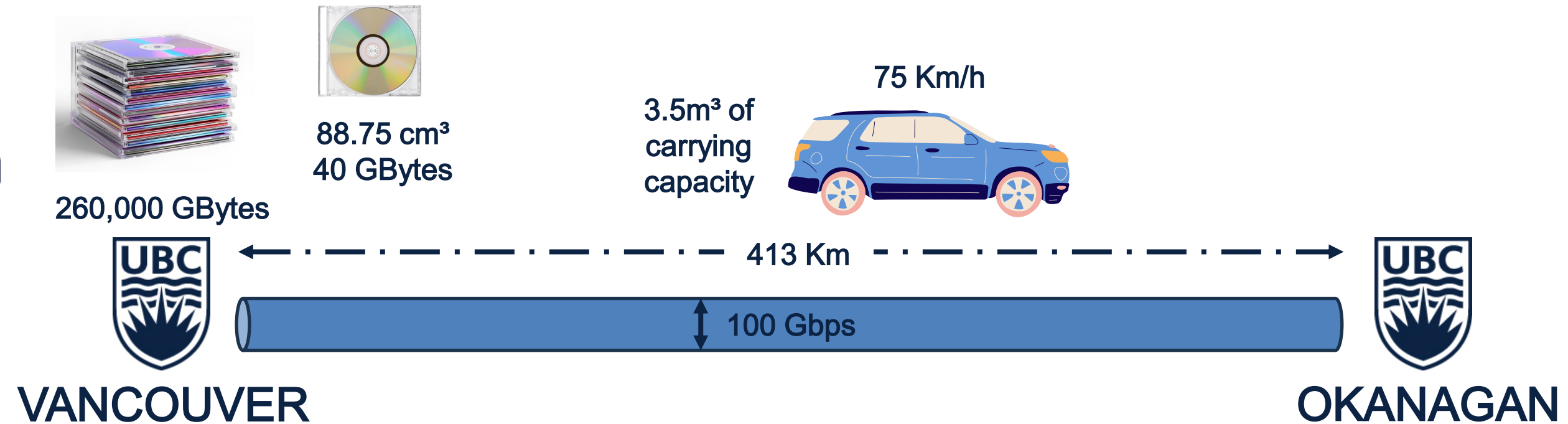
After 180 minutes, how many gigabytes of data would have arrived at UBC-O by SUV?

A. 0 GB ($413/75 = 5.51$ hours = 330.4 min $\Rightarrow$ 0 GB)

B. 135 000 GB

C. 141 646 GB

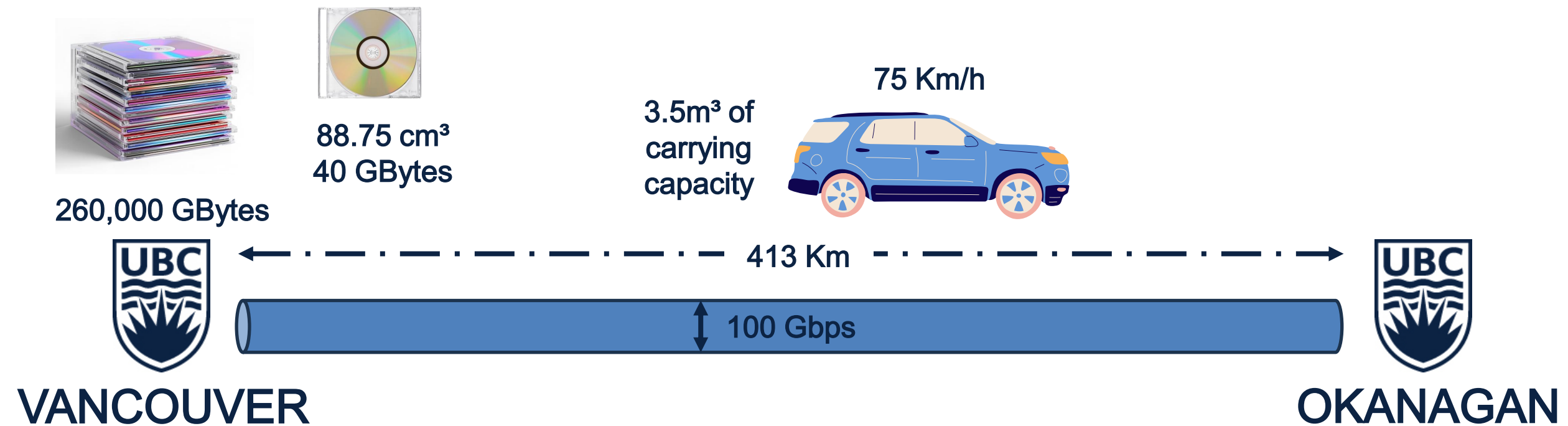
Clicker Question



Which will get **all** the data to UBC-O faster?

- A. SUV
- B. Network connection
- C. Insufficient data to determine

Clicker Answer



Which will get **all** the data to UBC-O faster?

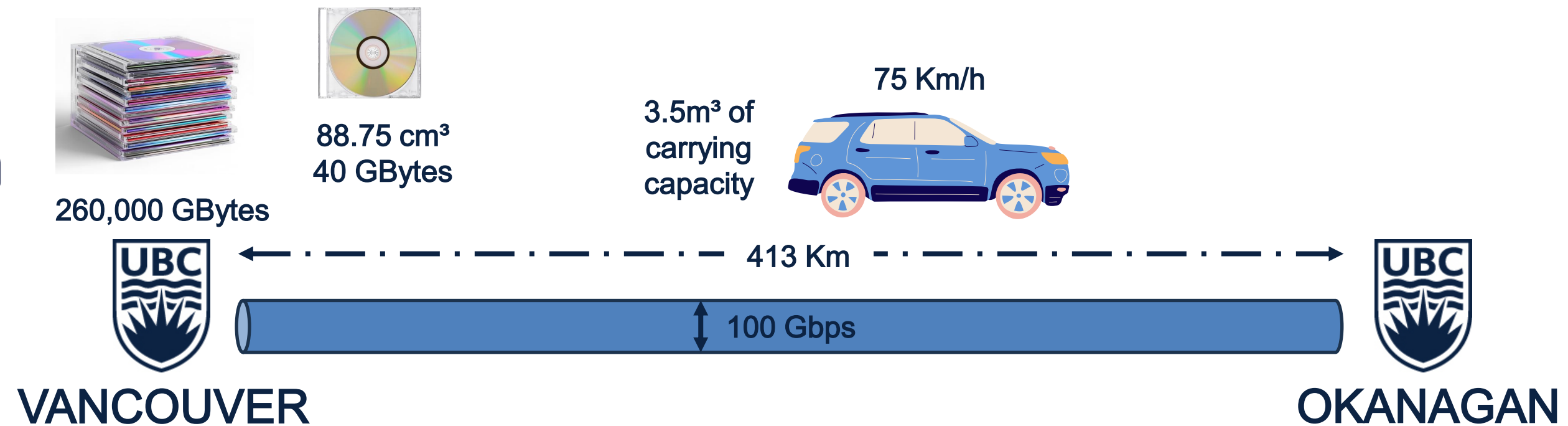
A. SUV takes $413/75 = 5.51$ hours = 330.4 min = 19 824 s

B. Data size = 260000 GB = $260000 * 8 * 2^{30}$ bits
Data rate = 100 Gbps = $100 * 10^9$ bps

Network takes $(260000 * 8 * 2^{30}) / (100 * 10^9)$ seconds
we divide by 3600 = **6.20 hours**

C. Insufficient data to determine

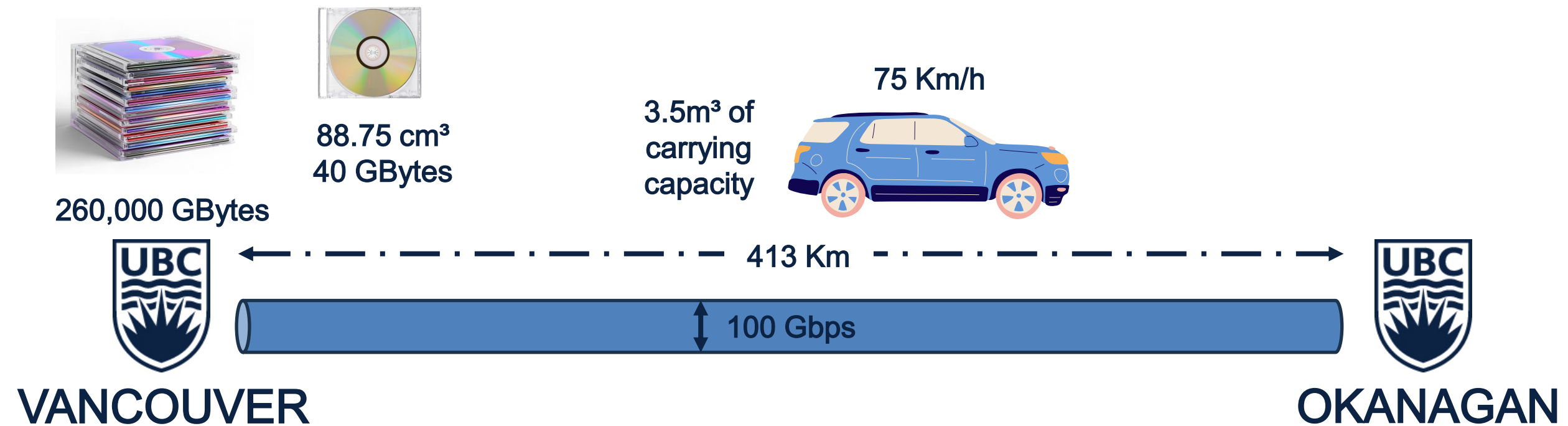
Clicker Question



What is the effective goodput of the SUV in this scenario?

- A. 104.9 Gbps
- B. 112.7 Gbps
- C. 683.5 Gbps

Clicker Answer



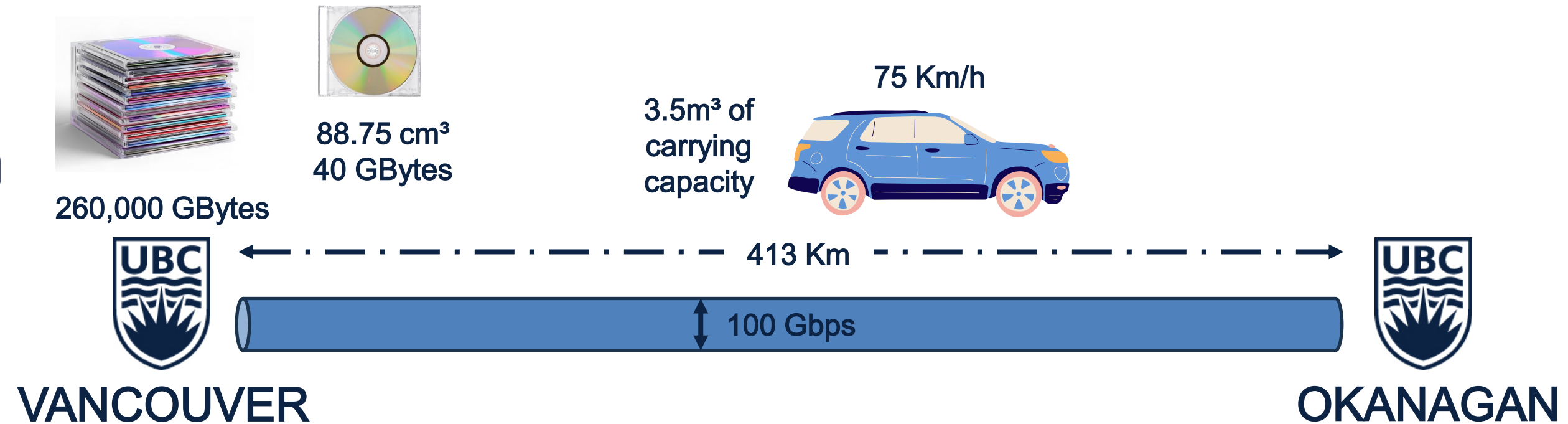
What is the effective goodput of the SUV in this scenario?

A. 104.9 Gbps

B. 112.7 Gbps $\left(\frac{260\,000 \times 8 \times 2^{30}}{19\,824} / 10^9 = 112.7 \text{ Gbps} \right)$

C. 683.5 Gbps

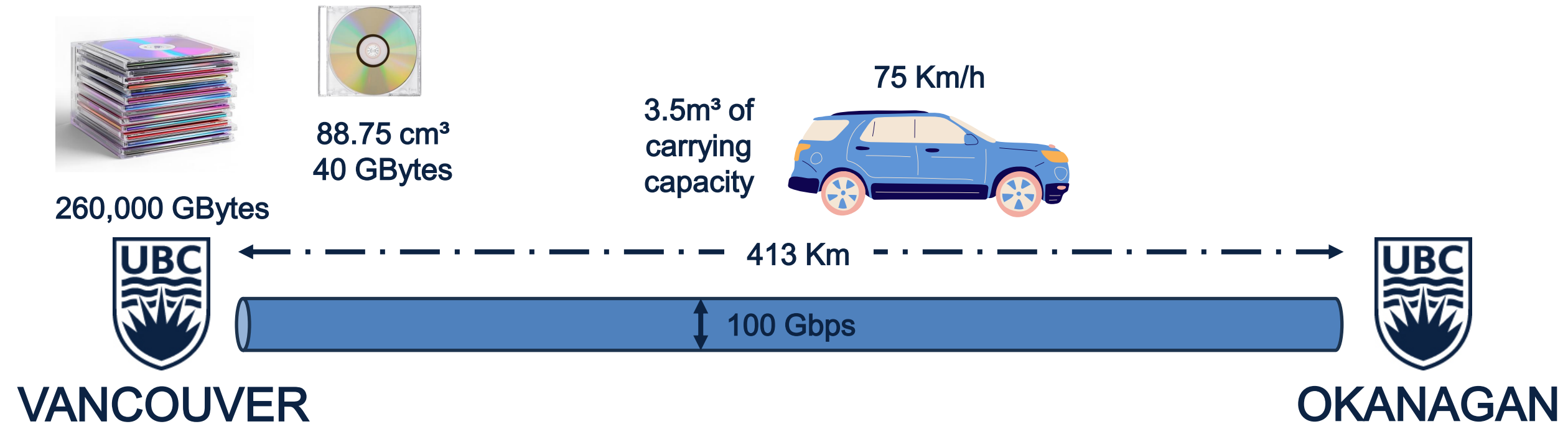
Clicker Question



Assuming the SUV's cargo area is filled with as many Blu-ray discs as possible, what is the maximum goodput of the SUV ?

- A. 340.2 Gbps
- B. 112.7 Gbps
- C. 683.5 Gbps

Clicker Answer



Assuming the SUV's cargo area is filled with as many Blu-ray discs as possible, what is the maximum goodput of the SUV ?

- A. 340.2 Gbps
- B. 112.7 Gbps

C. 683.5 Gbps

$$\frac{3.5 \times 10^6}{88.75} = 39\,436 \text{ discs}$$

$$\frac{39\,436 \times 40 \times 8 \times 2^{30}}{19\,824} / \times 10^9 = 683.5 \text{ Gbps}$$

Actually Used In Real Life

- Amazon snow family
 - Snowcone – up to 22TB – 4.5 lbs
 - <https://aws.amazon.com/snowcone/>
 - Snowball – up to 210TB – 50 lbs
 - <https://aws.amazon.com/snowball/>
 - Snowmobile – up to 100PB – 40 tons
 - <https://aws.amazon.com/snowmobile/>

Snowcone



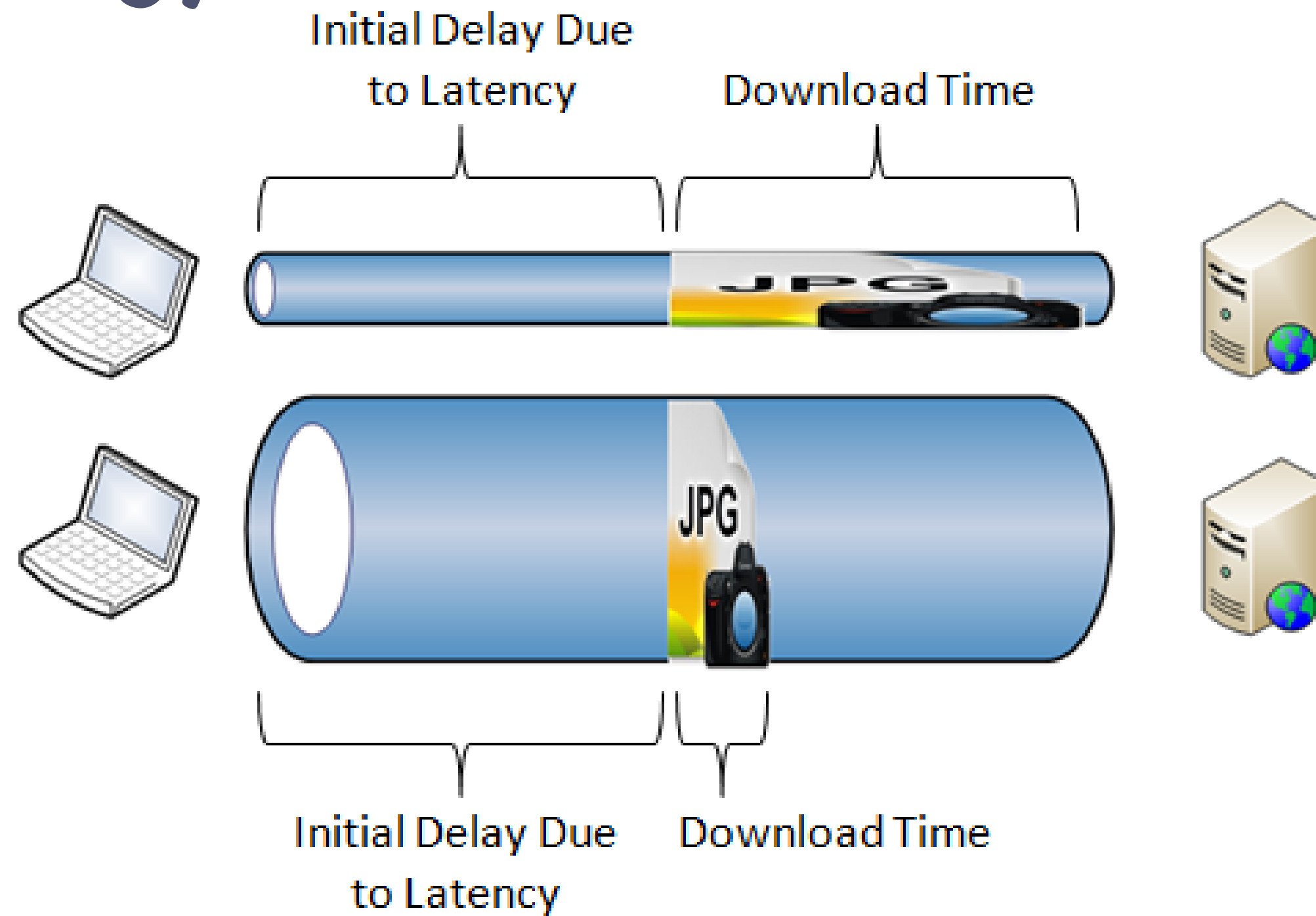
Snowball



Snowmobile



Pipe Analogy



Network Metrics: Latency

- **Latency:** delay from when something is sent until it is received
 - “Something” depends on context, but must be consistent
- Examples:
 - Packet latency: from start of sending packet until completely received
 - Bit/byte latency: from start of sending bit/byte until that bit/byte is completely received

Some facts about latency

- Fact One: Making more bandwidth is easy
- Fact Two: Once you have bad latency you're stuck with it
- Fact Three: Current consumer devices have appallingly bad latency
- Fact Four: Making limited bandwidth go further is easy
- This is not to say bandwidth is unimportant: Bandwidth Still Matters

<http://www.stuartcheshire.org/rants/latency.html>

Network Metrics: RTT

- **Round Trip Time (RTT):** latency for sending something and receiving something back
 - Latency for message + latency for response + processing time
 - The RTT for a bit is not the same as the RTT for a request and a response
- Easier to compute than one-way latency (single clock)
- Reported by ping, traceroute, etc.

Ping (measure RTT)

Observe that not all ping times are the same

```
$ ping -c 10 www.cs.ubc.ca
PING www.cs.ubc.ca (142.103.6.5): 56 data bytes
64 bytes from 142.103.6.5: icmp_seq=0 ttl=59 time=2.145 ms
64 bytes from 142.103.6.5: icmp_seq=1 ttl=59 time=2.177 ms
64 bytes from 142.103.6.5: icmp_seq=2 ttl=59 time=2.333 ms
64 bytes from 142.103.6.5: icmp_seq=3 ttl=59 time=2.353 ms
64 bytes from 142.103.6.5: icmp_seq=4 ttl=59 time=8.583 ms
64 bytes from 142.103.6.5: icmp_seq=5 ttl=59 time=8.446 ms
64 bytes from 142.103.6.5: icmp_seq=6 ttl=59 time=8.572 ms
64 bytes from 142.103.6.5: icmp_seq=7 ttl=59 time=8.452 ms
64 bytes from 142.103.6.5: icmp_seq=8 ttl=59 time=2.229 ms
64 bytes from 142.103.6.5: icmp_seq=9 ttl=59 time=2.221 ms

--- www.cs.ubc.ca ping statistics ---
10 packets transmitted, 10 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 2.145/4.751/8.583/3.073 ms
```

Network Metrics: Jitter

- Jitter: variation in latency and/or RTT



What Causes Jitter?

- Different paths for packets
- Network congestion
- Not implementing packet prioritization
- Poor hardware performance (old equipment)
- Wireless jitter (interference in medium)

In-class Activity

- ICA21

Next Topic: Network Performance - Delay